Session Objectives

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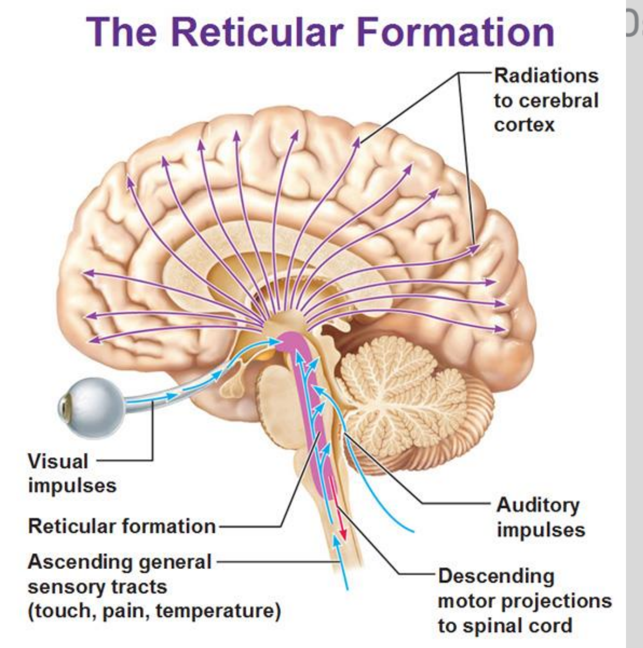
- Describe the anatomical organization and divisions of the reticular formation
- Understand the major functions of the reticular formation
- Decipher the neurochemistry of the reticular formation
- Master the neurophysiology of sleep
- Summarize the clinical implications of reticular formation dysfunction



See detailed objectives in CPG

***PLEASE READ THE LECTURE NOTES** (SRX bricks not available)*

Session Objectives Mapped to COMLEX Blueprint

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- **Dimension 2**
  - **Clinical presentation 4.** Patient presentations related to the nervous system and mental health
    - 4.12 Sleep disturbances
      - Sleep disorders, including obstructive sleep apnea, somnambulism, insomnia, excessive daytime sleepiness, sleep-wake disorders, narcolepsy, night terrors, parasomnias
      - cognitive impairments, including altered level of consciousness, mild cognitive impairment, amnesia, coma, confusion, delirium, disorientation, subcortical and cortical dementia (eg, Alzheimer disease, Huntington disease, Parkinson disease)

Source: https://www.nbome.org/Content/Exams/COMLEX-USA/COMLEX-USA_Master_Blueprint_2018-2019.pdf

Session Objectives Mapped to USMLE Blueprint

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V. Nervous System & Special Senses

A. Normal Processes

2. Organ structure and function

ii. Brain stem (eg, cranial nerves and nuclei, reticular formation, anatomy and blood supply, control of eye movements

B. Abnormal Processes: Health and Health Maintenance, Screening, Diagnosis, Management, Risks, Prognosis

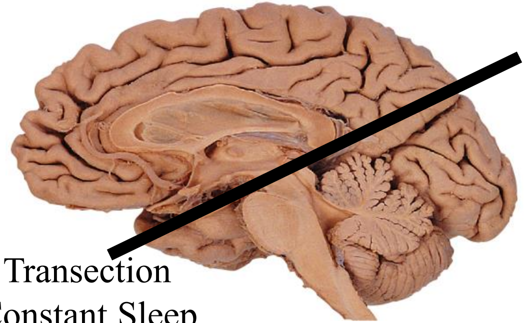
5. Cranial and peripheral nerve disorders

v. Global cerebral dysfunction**:** altered states of consciousness; delirium; coma/brain death

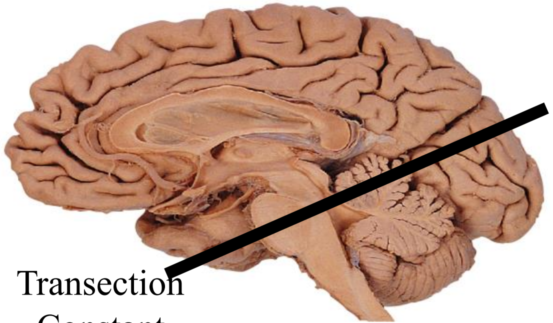
x. Sleep disorders: cataplexy and narcolepsy; circadian rhythm sleep-wake disorder; insomnia, primary; sleep terror disorder and sleepwalking; REM sleep behavior disorder; restless legs syndrome

Source: https://www.usmle.org/sites/default/files/2022-01/USMLE_Content_Outline_0.pdf

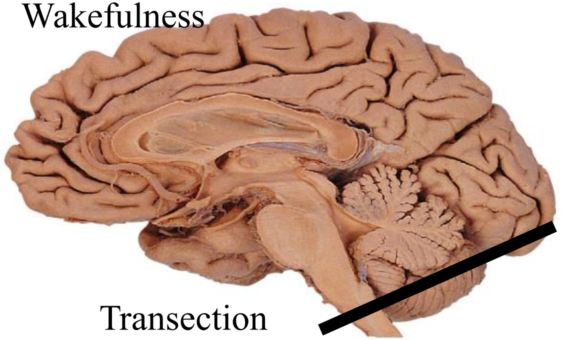
What is reticular formation?



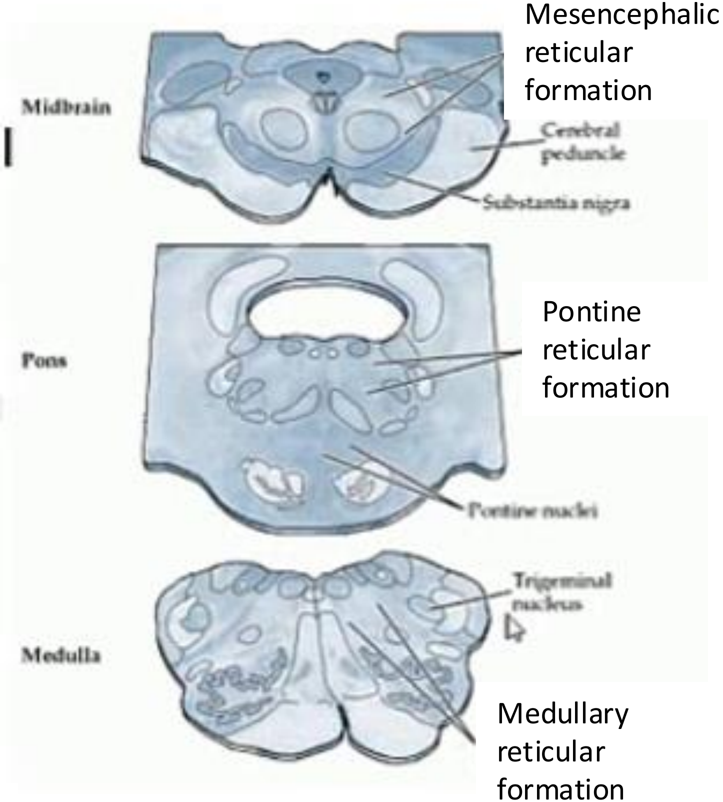
Transection
Constant Sleep



Transection
Constant
Wakefulness



Transection
Normal Sleep-Wake



Reticular activating system discovered in cats in 1949, Moruzzi and Magoun, *Electroencephalography and Clinical Neurophysiology*

What is the Reticular Formation?

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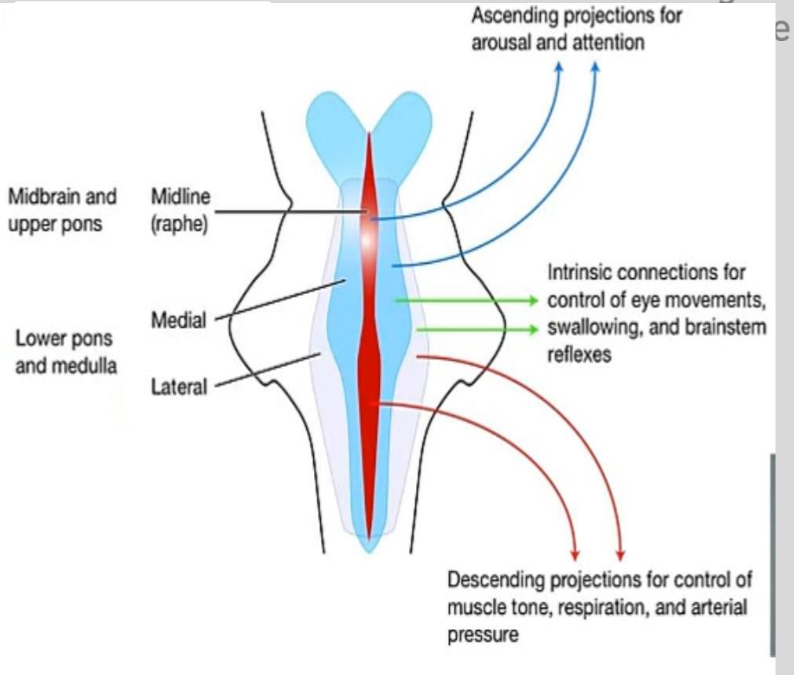
- **No definitive definition**
- Consists of
  - Extended rostrally to the lateral hypothalamus and subthalamic nuclei; Caudally to the spinal cord grey matter
  - Short **interneurons** for **brain stem reflexes**
  - Long **projection neurons** with cell bodies in the brain stem & hypothalamus and their axons **ascending** to the **forebrain** or **descending** to the **spinal cord**
- Do **not** consist the cranial nerve nuclei or the long tracks passing through the brain stem



Nuclei of the Reticular Formation

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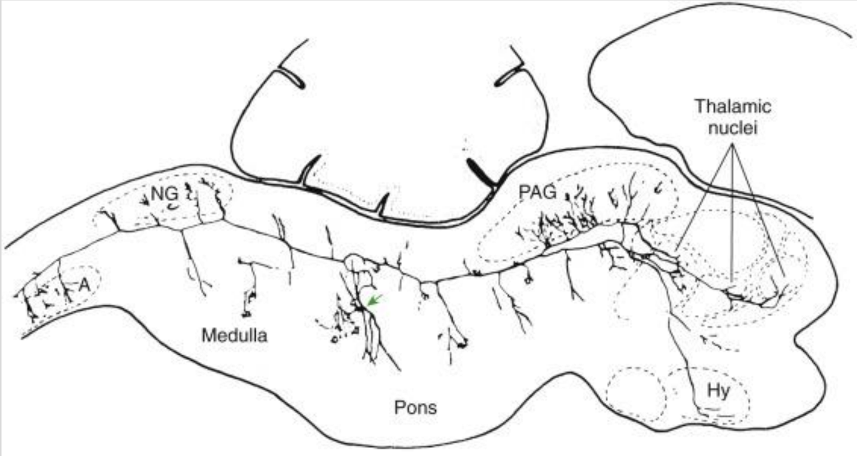
Projection neurons release the following neurotransmitters to send **diffuse, modulatory** inputs to cerebral cortex, thalamus, hypothalamus and spinal cord:

- **Serotonin** (5-HT)
- **Norepinephrine** (NE)/epinephrine
- **Dopamine** (DA)
- **Acetylcholine** (ACh)
- **Histamine**



Diffuse projection system:
It is estimated that a single neuron in this network may have synapses with as many as 25,000 other neurons.

Serotonergic System

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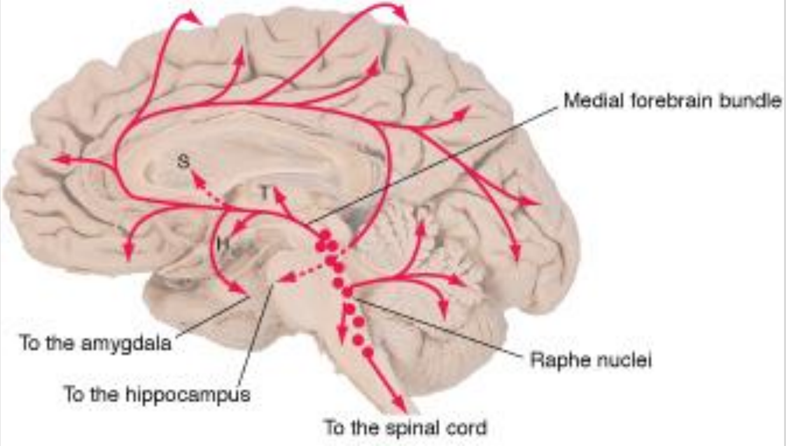
Cell bodies (nuclei):

- **Raphe nuclei:** along the entire brainstem
  - **dorsal raphe** nuclei (**midbrain**), project to the **forebrain**
  - **raphe magnus** nucleus (**medulla**), project to the **brainstem and spinal cord**

Functions:

- **Arousal** & **selective attention** (reticular activating system, **RAS**)
- **Sleep**
- Sensory processing, e.g., **pain**
- Regulation of muscle tone & segmental spinal reflexes, e.g., **reticulospinal system**



Noradrenergic System

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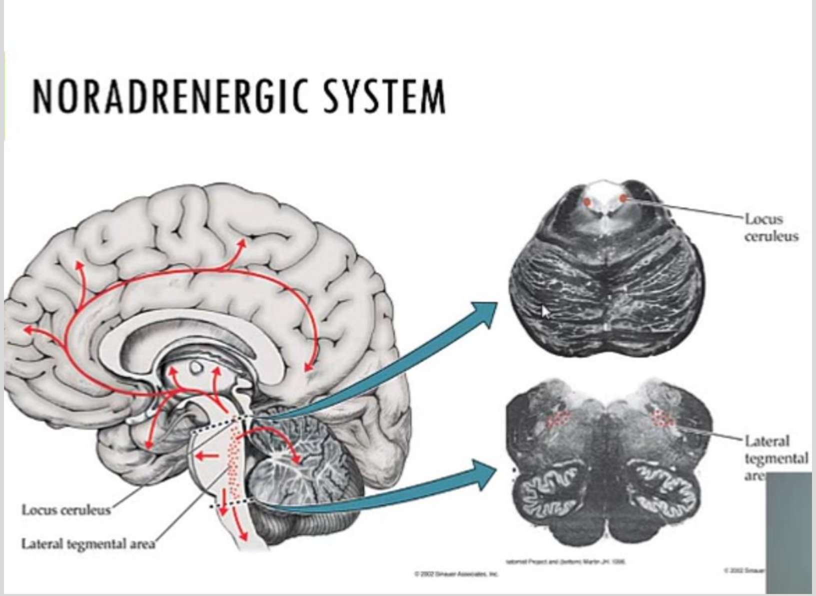
Cell bodies (nuclei):

- **Locus coeruleus**, project to all parts of the CNS
- Lateral tegmental area

Functions:

- **Arousal** & selective attention (reticular activating system, **RAS**)
- **Sleep**
- Sensory processing, e.g., **pain**
- **Autonomic** functions



Dopaminergic System

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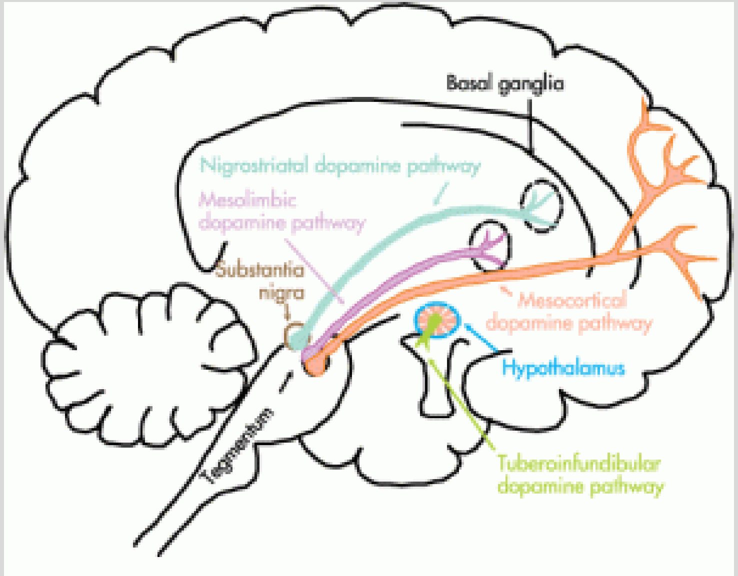
Pathways - Functions:

- Ventral tegmental area **(VTA),** mesencephalon
  - **Mesolimbic** – addiction, positive symptoms of schizophrenia, e.g., hallucinations
  - **Mesocortical** – cognitive deficits and hypokinesia in Parkinson's disease (PD), and "negative" symptoms of schizophrenia
- **Nigro-striatal system**
  - Originate in **substantia nigra pars compacta** – coordinating movements, major pathway related to PD
- **Tubero-infundibular system** - **Hypothalamus**, inhibit **prolactin** release from pituitary gland



Cholinergic System

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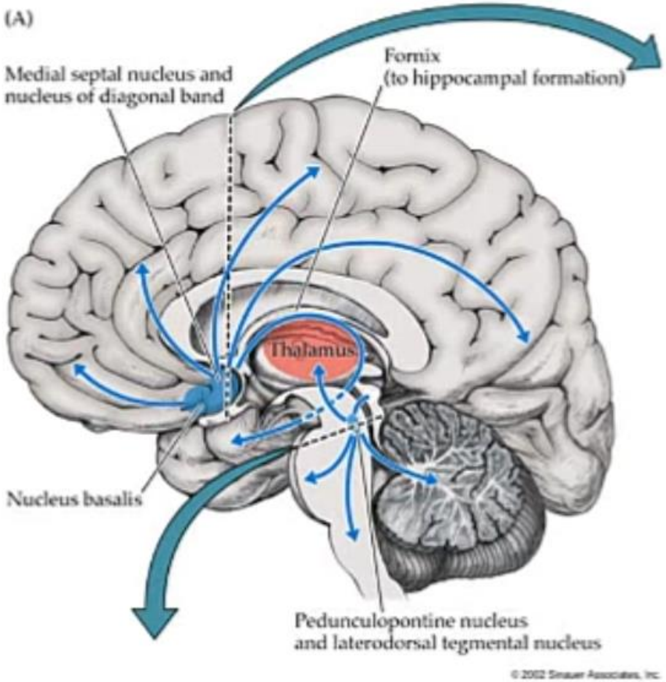
Cell bodies:

- **Nucleus basalis** of **Meynert** (**NBM**, **basal forebrain** system), project to the entire cortex
- **Pedunculopontine tegmental nuclei** (PPT), project to brainstem and spinal cord
- NBM & PPT activate cortical neurons via thalamic neurons

Functions:

- **Learning & memory, cognition - NBM (Alzheimer's Disease, Parkinson's Disease)**
- **Arousal** (**RAS**) and selective attention
- **Locomotion** - **PPT** (**fixed gait in PD**)



Histaminergic System

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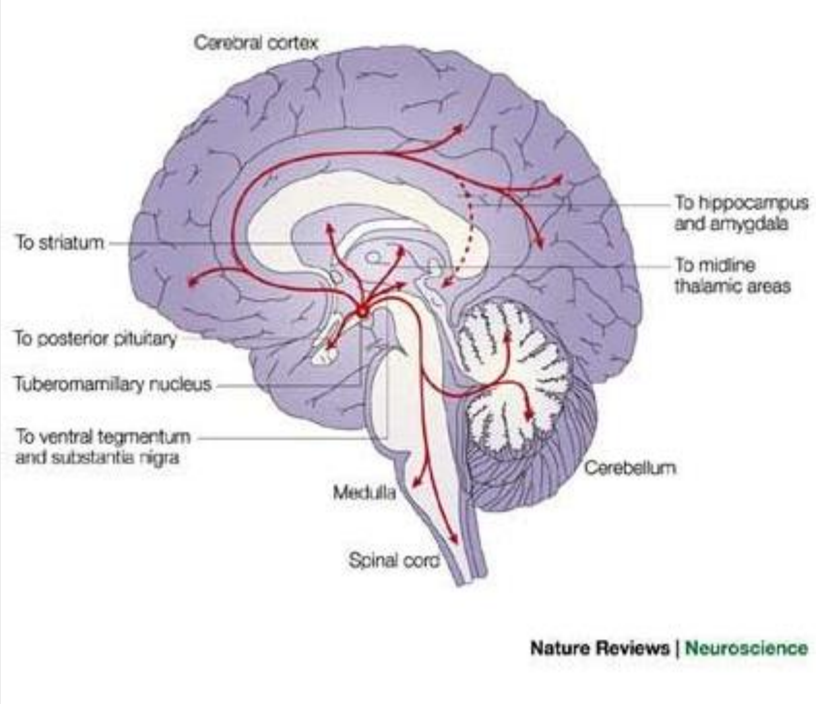
Cell bodies:

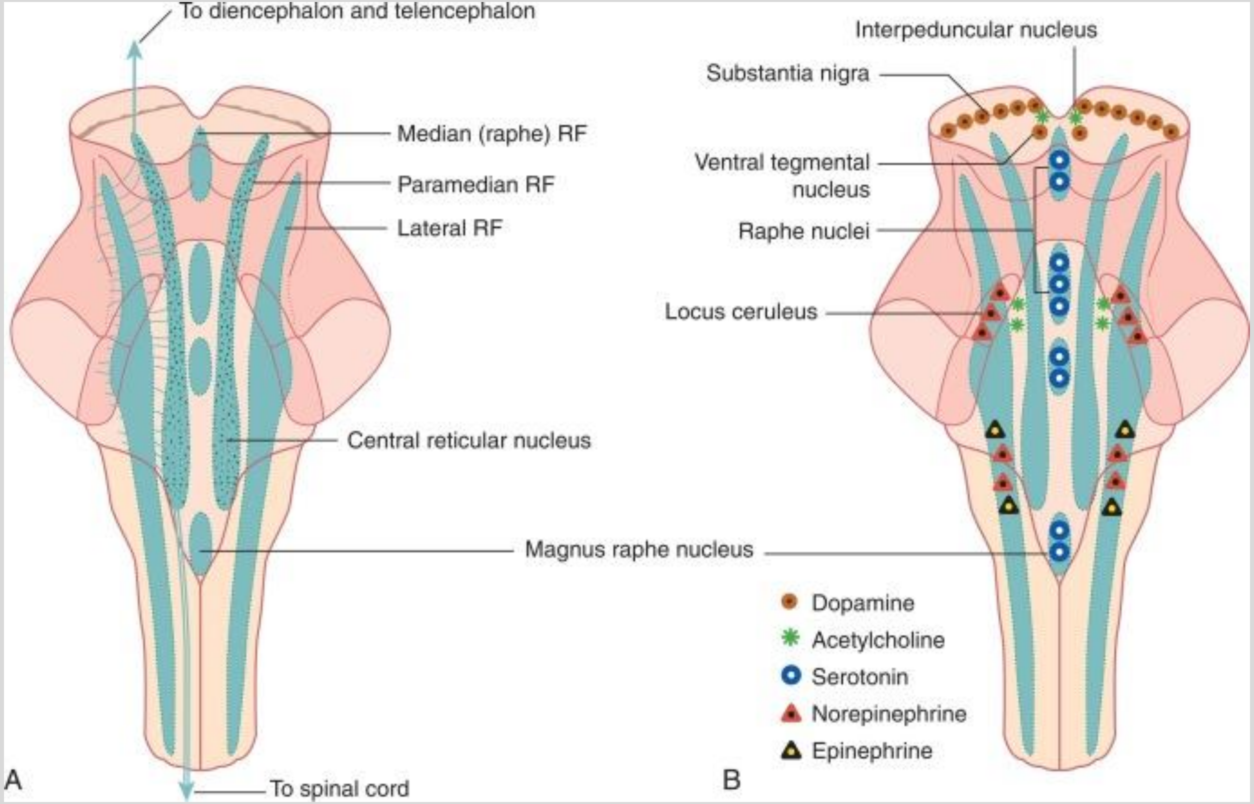
- **Posterior lateral hypothalamus** and **tuberomammillary** nucleus, project to all parts of CNS

Functions:

- **Arousal** (**RAS**)
- **Sleep**

(Think about what Benadryl can do)





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Functions of Reticular Formation

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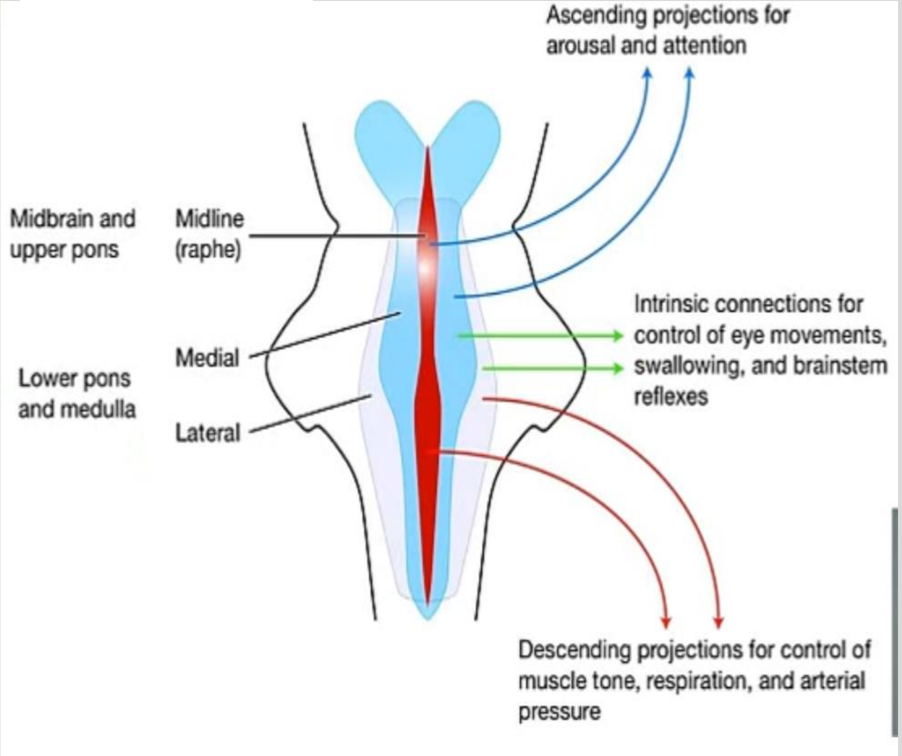
Ascending:

- **Arousal**
- **Attention**
- **Sleep**

Descending:

- **Motor Functions**
- **Pain**
- **Micturition**
- **Cardiac Reflexes**
- **Respiration**



Functions of Reticular Formation

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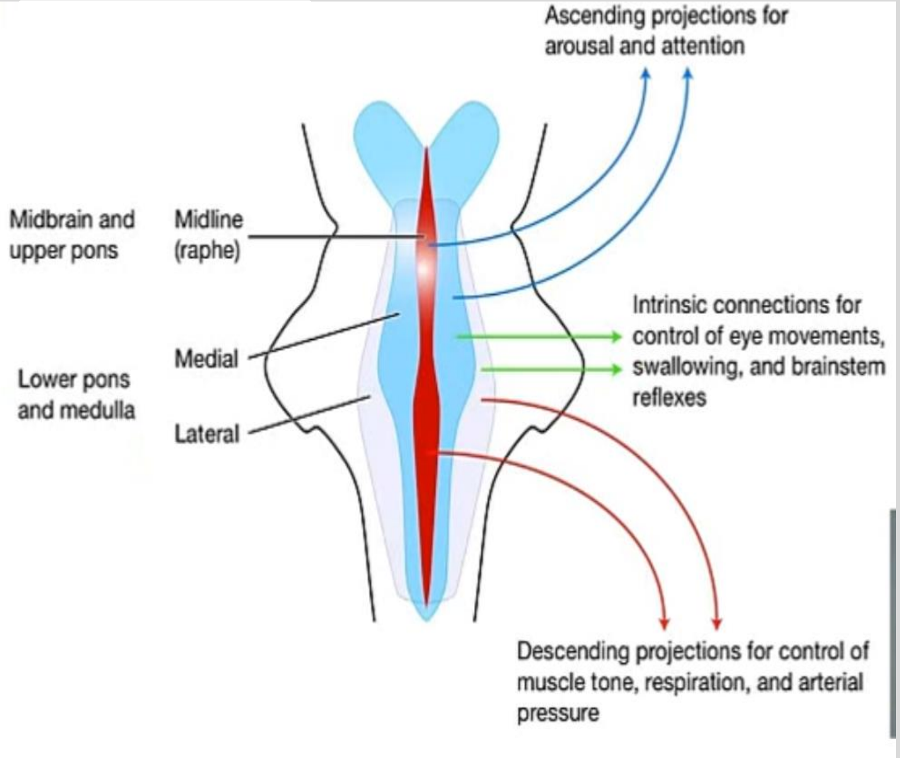

Traumatic Brain Injury

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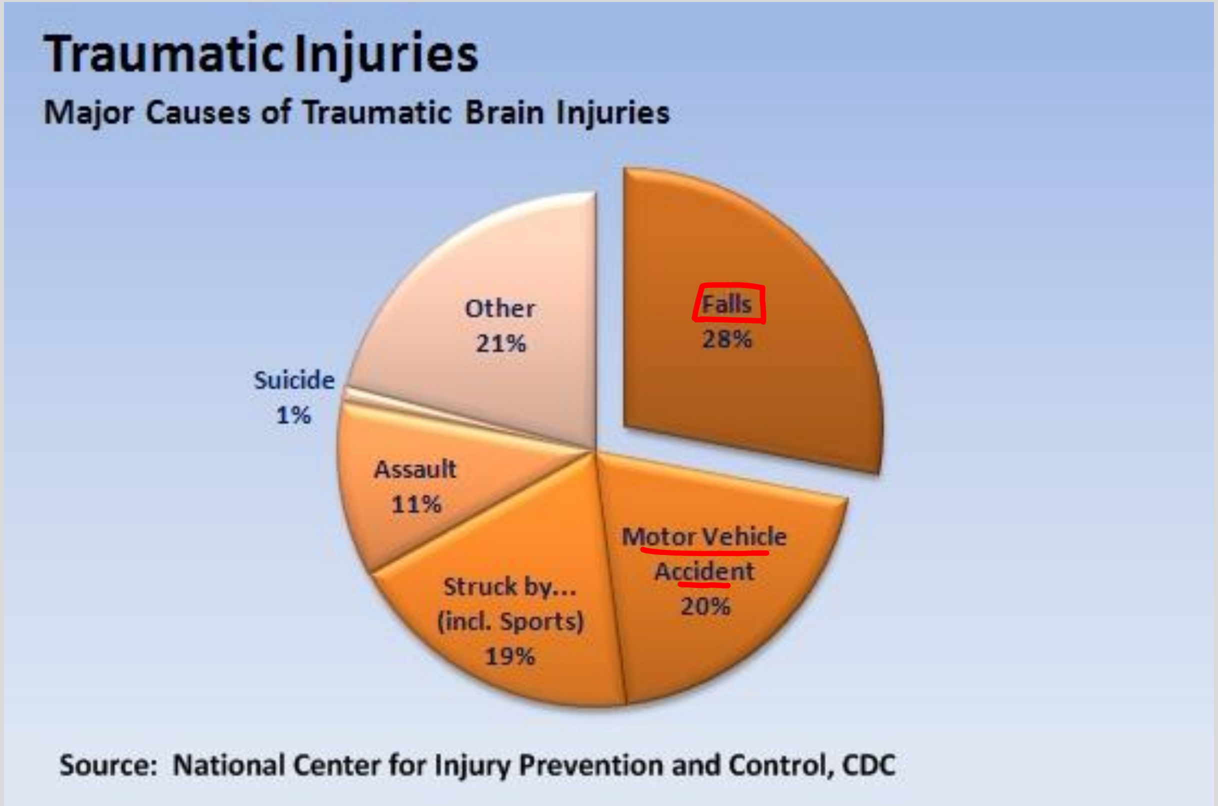

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GLASGOW COMA SCALE

Eye Opening

Spontaneous	4
To loud voice	3
To Pain	2
None	1

Verbal Response

Oriented	5
Confused/Disoriented	4
Inappropriate words	3
Incomprehensible sounds	2
None	1

Best Motor Response

Obeys	6
Localizes	5
Withdraws (flexion)	4
Abnormal flexion	3
Extension	2
None	1

*Normal Score = 15

*Mild Head injury = 14-15

*Moderate Head injury = 9-13

*Severe Head injury = less than or equal to **8**

Lowest score = 3 (not 0)

Persistent vegetative state

("unresponsive wakefulness") is marked by **moderate arousal** and **low awareness**.

- **Respond to external stimuli**, such as food and touch
- Eyes open and close
- Intact sleep-wake cycle
- Capable of swallowing
- Sometimes show very simple behaviors, teeth grinding, eye tracking, but do not show external evidence of consciousness
- Do **not** need artificial respiration
- Rarely return to normal consciousness if prolonged by more than a year

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Terry Schiavo (1963-2005)

Additional brain processing during vegetative state revealed by fMRI

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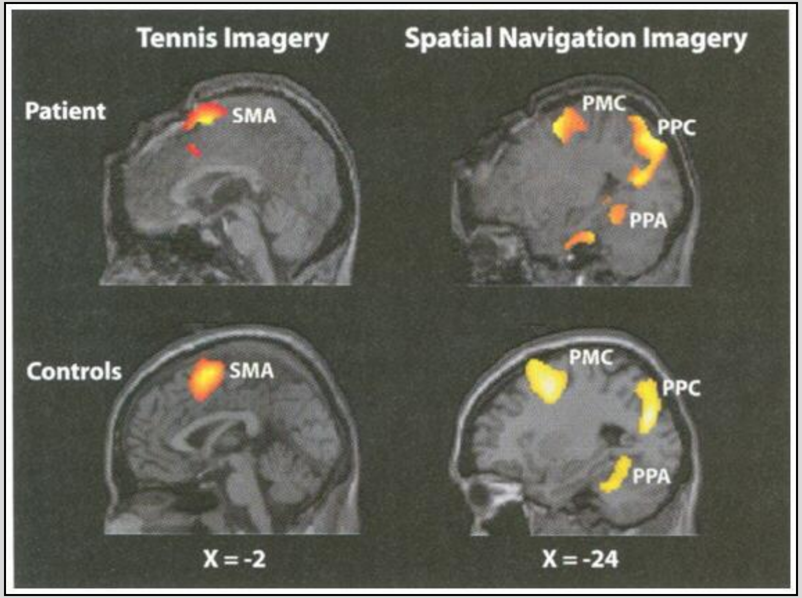


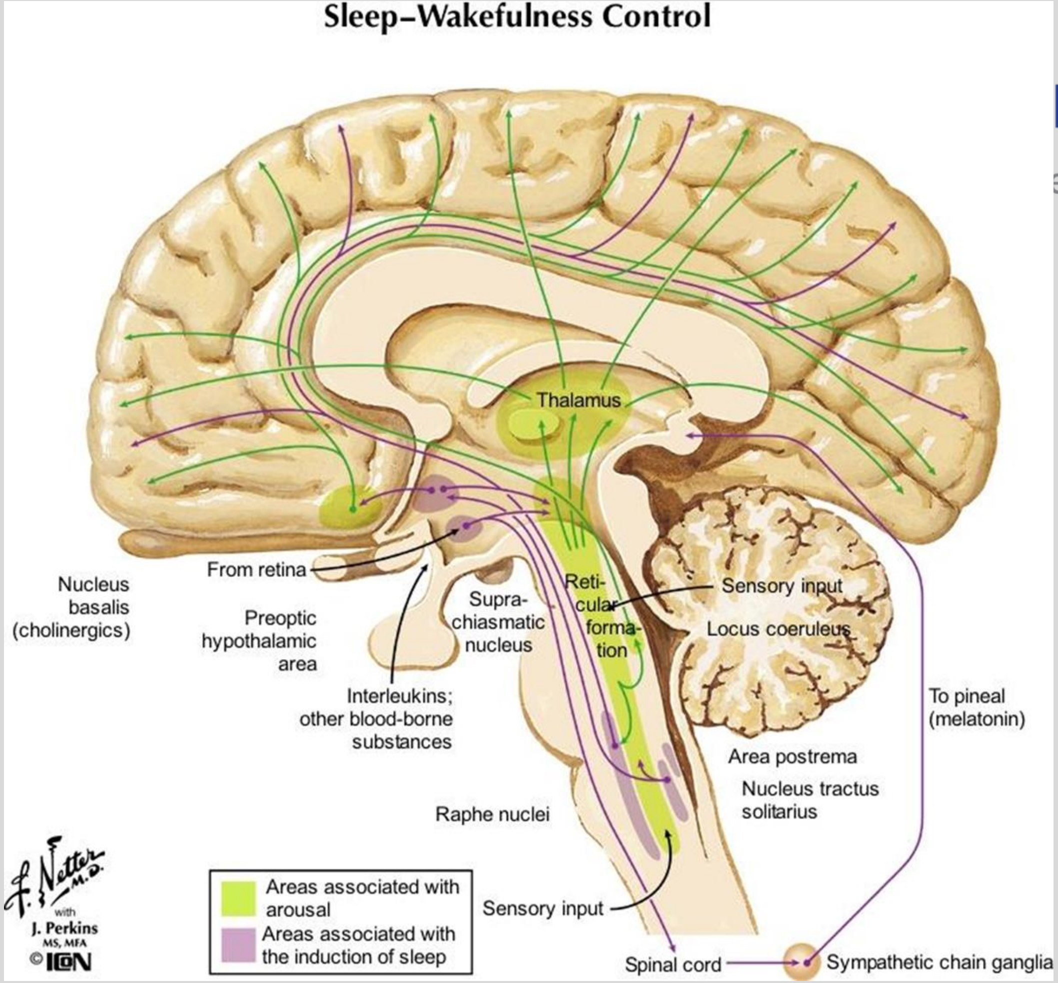
Detecting Awareness in the Vegetative State. Owen, Adrian; Coleman, Martin; Boly, Melanie; Davis, Matthew; Laureys, Steven; Pickard, John

Science. 313(5792):1402, September 8, 2006.

Fig. 1 BOLD imaging (to measure blood flow) showed activation of different brain areas when the patient was asked to imagine different types of activities. When the patient was asked to imagine playing tennis, supplementary motor area (SMA) lit up; in contrast, when the patient was asked to imagine going around the house, the parahippocampal gyrus (PPA), posterior parietal-lobe (PPC), and lateral premotor cortex (PMC) lit up. These pattens were similar to healthy volunteers, suggesting this vegetative state patient maintained awareness.

Definition of sleep:

- Reduced motor activity
- Decreased response to stimulation
- Stereotypic postures
- Relatively easy reversibility



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EEG is associated with different stages of sleep and wakefulness

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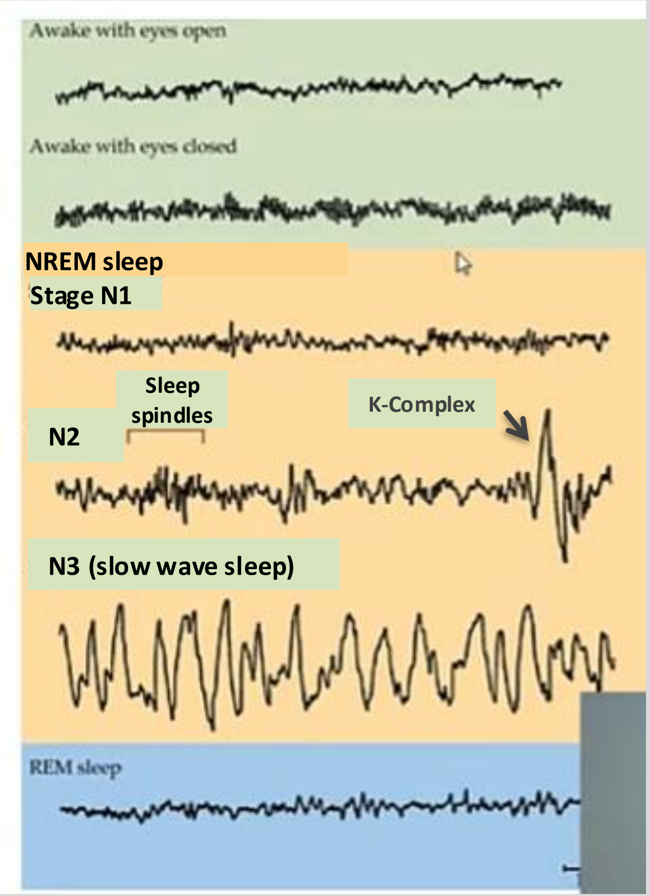
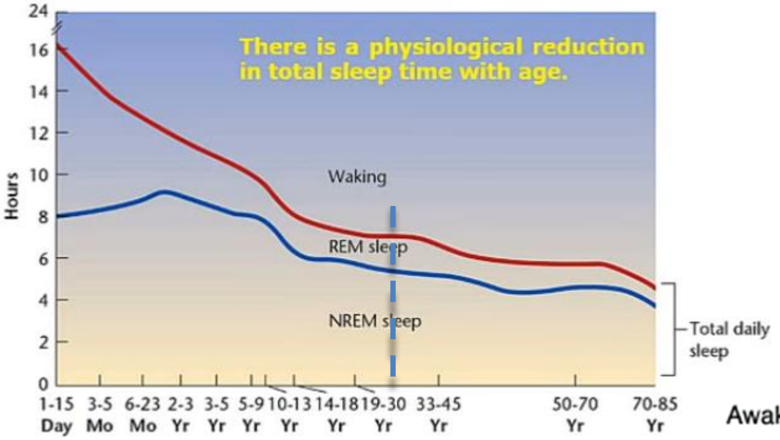
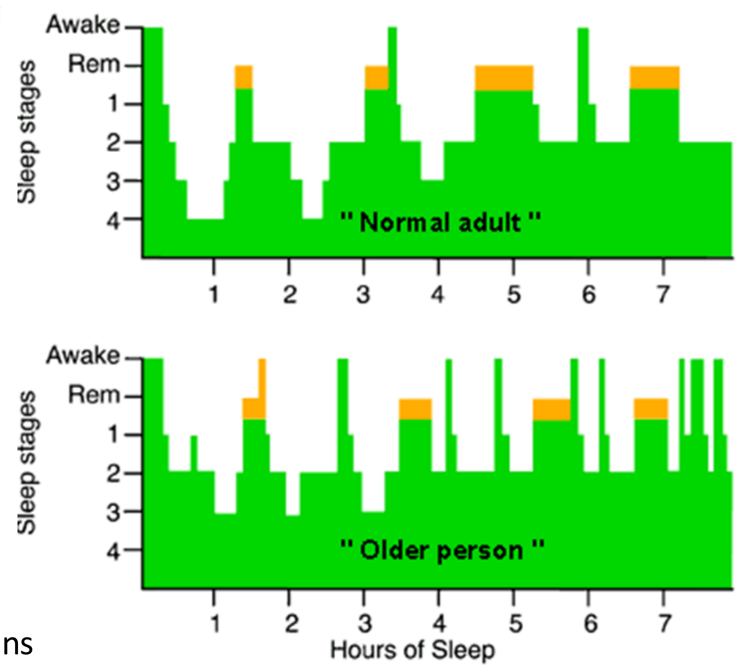


- **Awake**
  - **Awake with eyes open:** high frequency, low amplitude **beta (β)** waves
  - **Relaxed wakefulness (eyes shut**) shows rhythmic 8–12-Hz **alpha (α)** waves
- **NREM (Non-REM) sleep**
  - Stage N1 shows mixed frequencies, especially 3–7-Hz **theta (θ)** waves
  - Stage N2 shows 12–14-Hz **sleep spindles** and **K-complexes**
  - Stage N3 **(slow wave sleep)** shows large-amplitude (>75 µV) 0.5–2-Hz **delta (δ)** waves.
- REM **(Rapid Eye Movement) sleep**
  - shows low-amplitude, mixed frequencies with saw-tooth waves, **close to beta (β)** waves

Changes in sleep cycle with age



Hypnogram



Old stage definitions

Neurochemistry of Sleep

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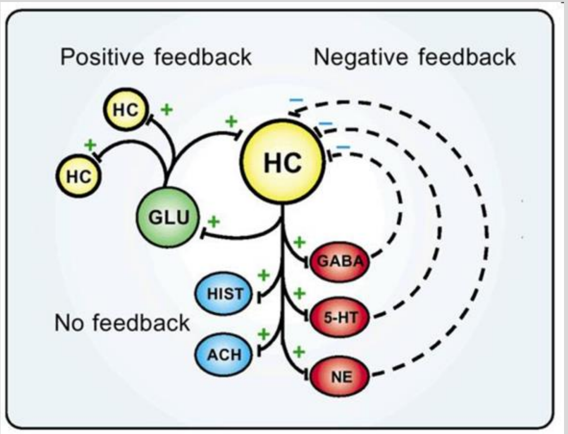
- Neurotransmitters promoting awakefulness:
  - Glutamate: universal
  - Serotonin neurons: *Raphe nuclei*
  - Norepinephrine neurons: *Locus coeruleus*
  - Acetylcholine neurons: *basal forebrain and dorsal tegmentum*
  - Histamine neurons: *Tuberomammillary nucleus*
  - **Hypocretin (Orexin)** neurons: ***Lateral hypothalamus***
    - Narcolepsy: low hypocretin in the brain



Neurochemistry of Sleep

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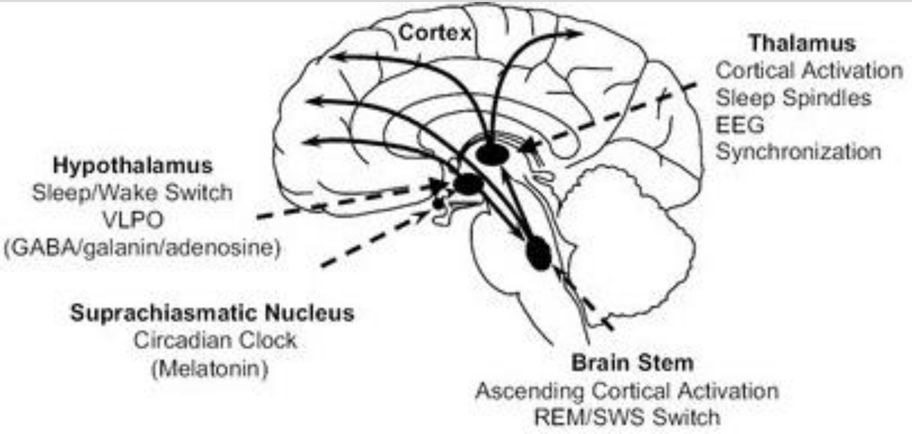
- Neurotransmitters promoting sleep:
  - Gamma-aminobutyric acid (**GABA**): *basal forebrain (medial septum and diagonal band) and hypothalamus (preoptic area); e.g., benzos, barbiturates*
  - **Melatonin**: *suprachiasmatic nucleus* → *pineal gland*



VLPO: ventrolateral preoptic nucleus

https://www.researchgate.net/publication/230863649_Future_Research_Directions_in_Sleep_and_ADHD_Report_of_a_Consensus_Working_Group/figures?lo=1

Narcolepsy Treatment

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Stimulants: to treat excessive daytime sleepiness

- Dextroamphetamine – sympathemimetic: promoting release and inhibiting reuptake of DA, NE, 5-HT
- Methylphenidate – NDRI (norepinephrine/dopamine reuptake inhibitor) → inhibits reuptake of NE and DA

Drug used as sleep aids

	Zaleplon (Sonata®)	Potentiate GABAs ability to open the $GABA_A$ chloride channel (increase channel opening frequency) very short acting	Sedation, Dizziness
	Zolpidem (Ambien®)	Potentiate GABAs ability to open the $GABA_A$ chloride channel (increase channel opening frequency) short acting	Sedation, Dizziness
	Eszopiclone (Lunesta®)	Potentiate GABAs ability to open the $GABA_A$ chloride channel (increase channel opening frequency) long acting	Sedation, Dizziness, Somnolence
	Chloral hydrate	Increase duration of $GABA_A$ chloride channel openings, at high concentrations can open directly	Sudden acute intoxication-death, Dependence, Severe withdrawal, Liver damage
	Buspirone (BuSpar®)	5-HT1A receptor partial agonist	Sedation, Ataxia, Should not be taken with MAOIs, Withdrawal
	Suvorexant (Belsomra®)	Orexin receptor antagonist	Sedation, mental cloudiness, CYP3A4
	Ramelteon (Rozerem®)	Melatonin MT1 and MT2 receptor agonist	hyperprolactinaemia, Possible teratogen, uses CYP1A2, CYP3A4, CYP 2C9

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Functions of Reticular Formation

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Ascending:

- **Arousal**
- **Attention**
- **Sleep**

Descending:

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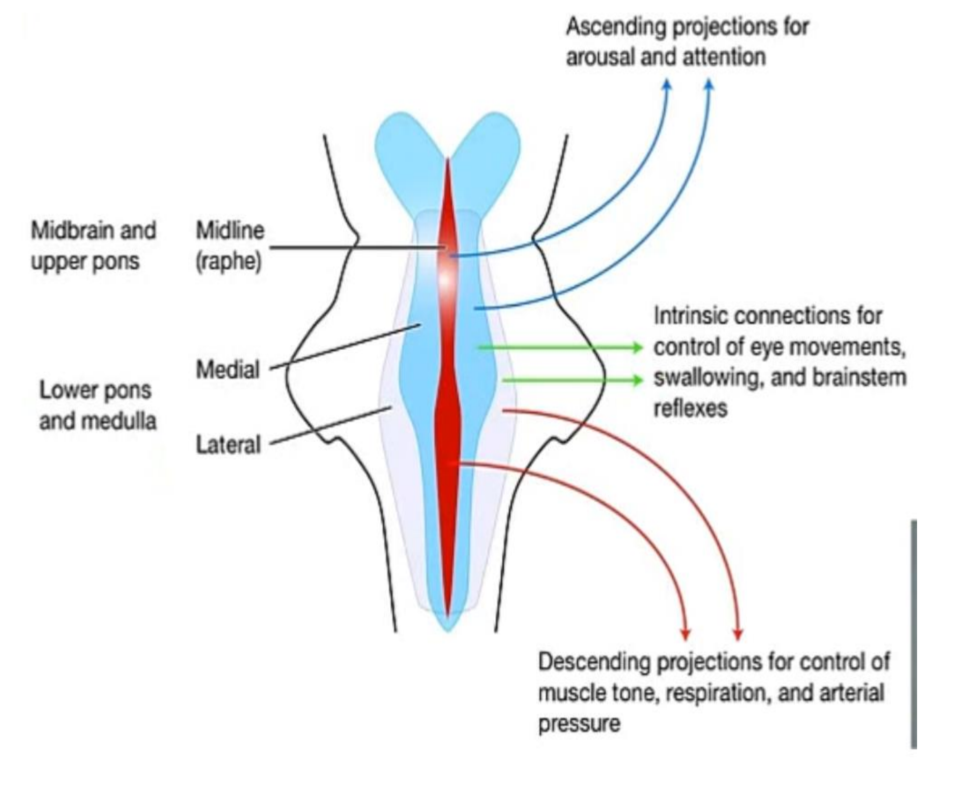

Descending Motor Control

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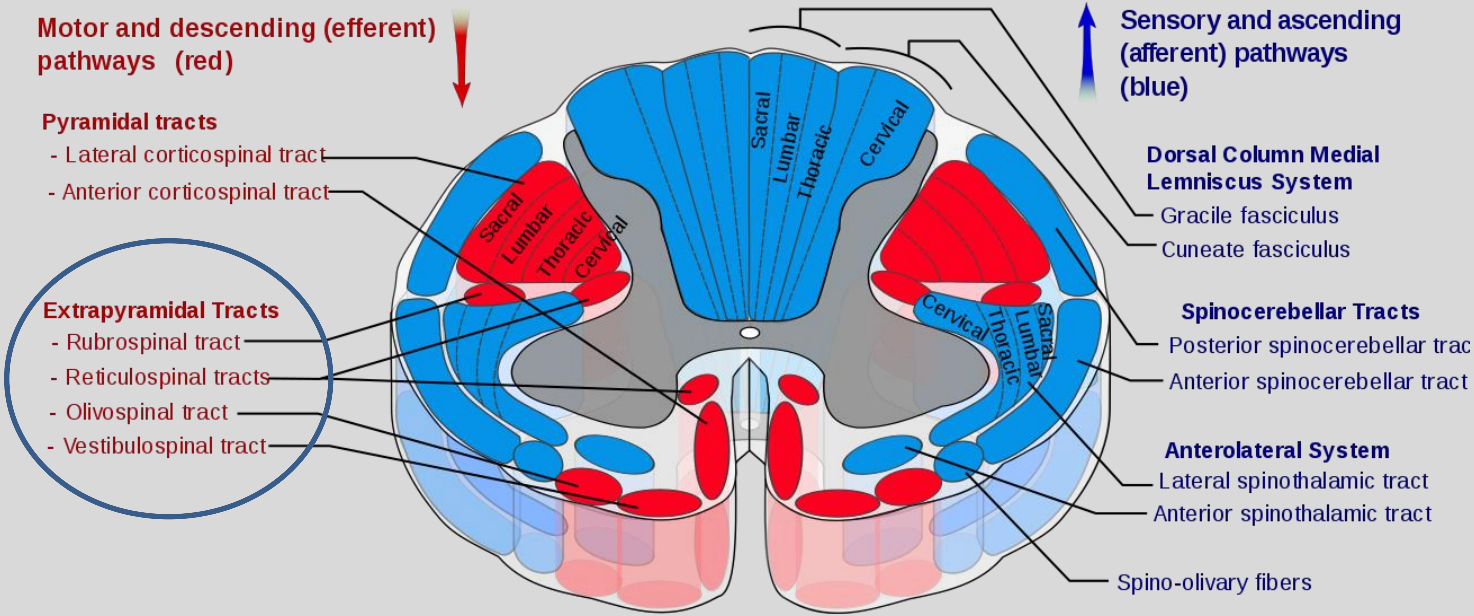

Chronic Pain Therapeutics

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Chronic pain patients - diminished descending pain inhibition

Current first line treatments for chronic pain engage descending pain inhibitory system:

- Amine antidepressants:
  - TCAs (tricyclic antidepressants):
    - Amitriptyline - serotonin norepinephrine reuptake inhibitor (SSNRI);
    - Desipramine: norepinephrine-selective reuptake inhibitor (NSRI), weaker therapeutic effect
    - Floxetine: selective serotonin reuptake inhibitor (SSRI), high dose can treat fibromyoalgia
  - Duloxetine: SNRI

The efficiency of conditioned pain modulation (CPM, "pain inhibits pain") predicts the efficacy of duloxetine.

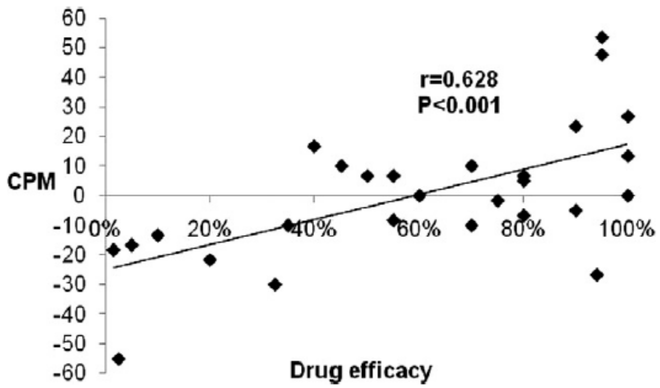
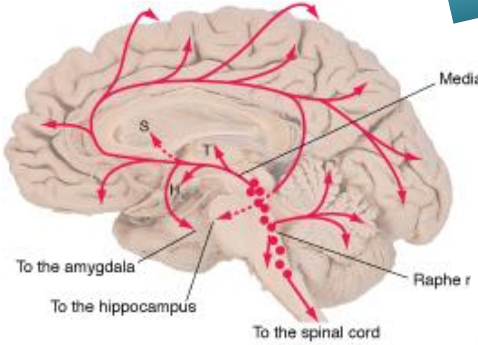


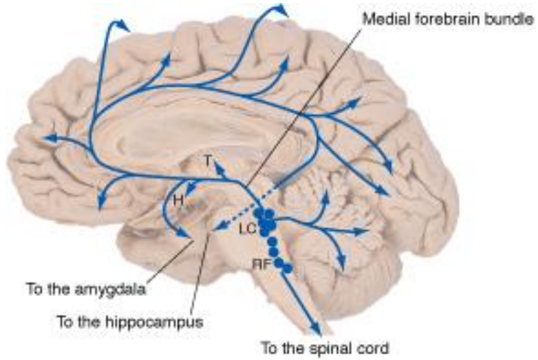
Fig. 1. Drug efficacy and pretreatment CPM. Patients with less efficient CPM (positive scores) reported higher drug efficacy and vice versa.

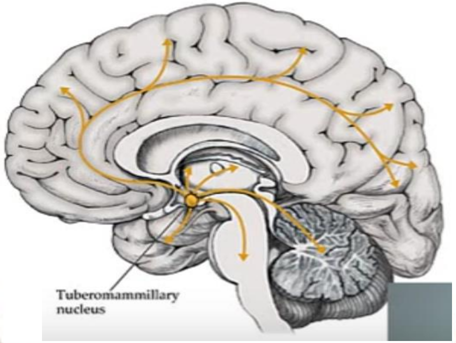
Yarnitsky, D., et al. (2012), Pain **153**(6): 1193-1198

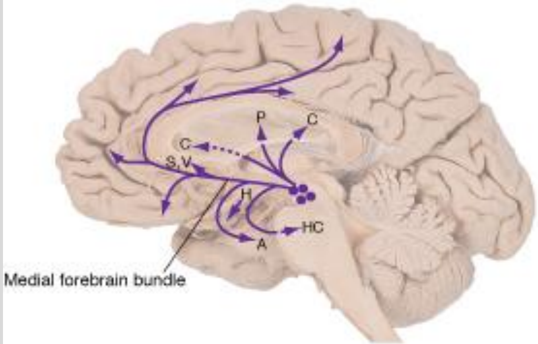
Summary

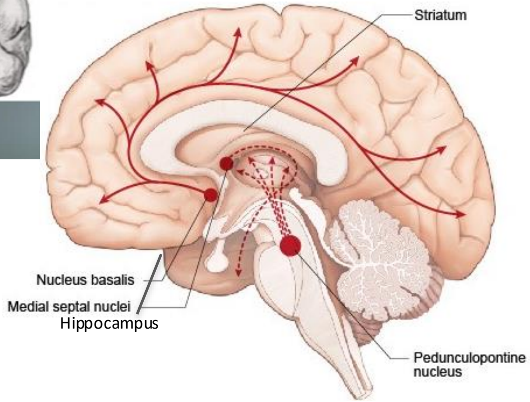
Which pathway?
Neurotransmitter?











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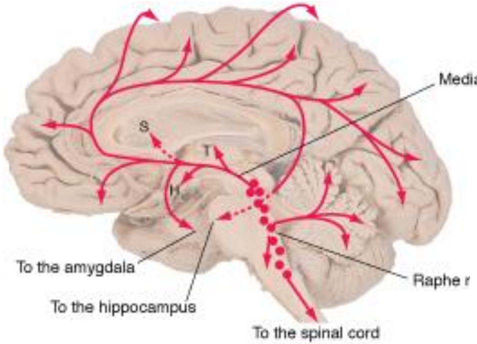
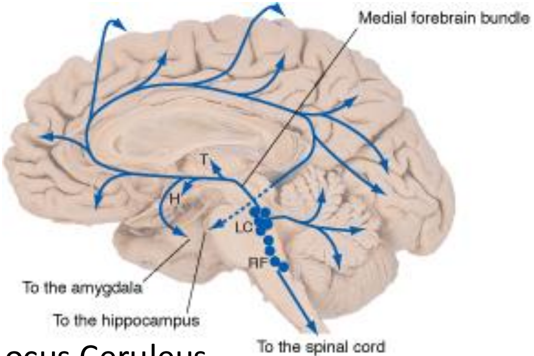
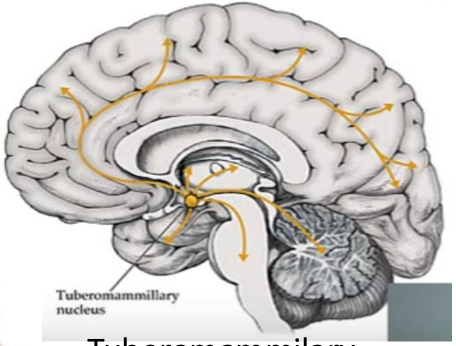
Summary



Raphe Nuclei (5-HT)

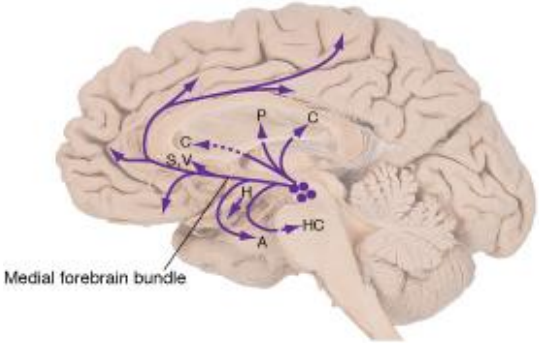


Locus Ceruleus
(NE)

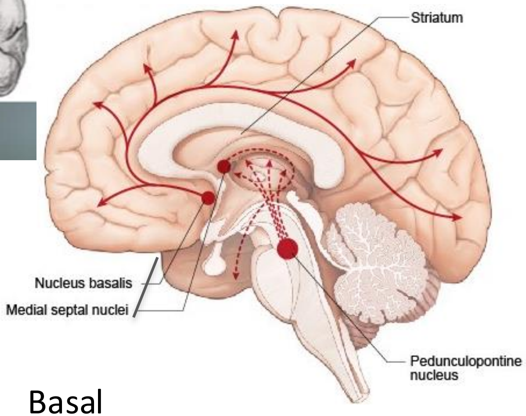


Tuberomammilary nucleus

(His)



Ventral tegmental &
Substantia Nigra
(DA)



Basal forebrain/Pedunculopontine
(ACh)

Summary

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Functions	Anatomical area	Neurotransmitter(s) (incomplete list)
Arousal, attention	Ascending reticular activating system (ARAS)	ACh, NE, DA, 5-HT, His
Sleep	Reticular formation, hypothalamus	ACh, His, NE, 5-HT, GABA, hypocretin
Motor control	Pedunculopontine tegmental nuclei (PPT), red nucleus, extrapyramidal tracts	ACh, DA, Glu, GABA
Pain modulation	Periaqueductal gray (PAG), Raphe magnus, locus coerulus	Enkephalin, 5-HT, NE
Bladder control	Pontine micturition center, PAG	Glu, GABA
Respiration	Medial parabrachial nucleus (breathing pattern)	Glu